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function_ref: a non-owning reference to a Callable

Abstract

This paper proposes the addition of `function_ref<R(Args...)>` to the Standard Library, a “*vocabulary type*” for non-owning references to `Callable` objects.

Changelog and polls

From P0792R1 to P0792R2

Changes

- Made *copy constructor* and *copy assignment* = `default`;
- Changed uses of `std::decay_t` to `std::remove_cvref_t`;
- Added “exposition only” `void*` and *pointer to function* data members;
- Moved “*Open questions*” section to “*Annex: previously open questions*”;
- Change `function_ref(F&&)` constructor’s precondition to use `remove_cvref_t` to check if `F` is an instance of the `function` class template;

- Dropped `function_ref<Signature>::` qualification in member function specification.

Polls

We want to prevent construction of `std::function` from `std::function_ref` (but not other callable-taking things like `std::bind`).

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We want to revise the paper to include discussion of ref-qualified callables.

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0 3 6 6 0

Forward paper as-is to LWG for C++20?

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3 9 3 0 0

Open questions

- Should the `!f` precondition when constructing `function_ref` from an instance `f` of `std::function` be removed? The behavior in that case is well-defined, as `f` is guaranteed to throw on invocation.
- The `std::is_nothrow_invocable` constraint in `function_ref` construction/assignment for noexcept signatures prevents users from providing a non-noexcept function, even if they know that it cannot ever throw (e.g. C functions). Should this constraint be removed? Should an explicit constructor without the constraint be provided?
- Propagating `const` to `function_ref::operator()` doesn't make sense when looking at `function_ref` as a simple "reference" class. `const` instances of `function_ref` should be able to invoke a mutable lambda, as the state of `function_ref` itself doesn't change. E.g.

```
auto l0 = []() mutable { };
const function_ref<void()> fr{l0};

fr(); // Currently a compilation error
```

An alternative is to only propagate `noexcept` from the signature to `function_ref::operator()`, and unconditionally `const-qualify` `function_ref::operator()`. Do we want this?

- We want to avoid double indirection when a `function_ref` instance is initialized with a `reference_wrapper`. `function_ref` could just copy the pointer stored inside the `reference_wrapper` instead of pointing to the wrapper itself. This cannot be covered by the *as-if* rule as it changes program semantics. E.g.

```
auto l0 = []{ };
auto l1 = []{ };
auto rw = std::ref(l0);

function_ref<void()> fr{rw};
fr(); // Invokes `l0`

rw = l1;
fr(); // What is invoked?
```

Is adding wording to handle `std::reference_wrapper` as a special case desirable?

- Is it possible and desirable to remove `function_ref`'s template assignment operator from F&& and rely on an implicit conversion to `function_ref` + the default copy assignment operator?
- Should `function_ref` only store a `void*` pointer for the callable object, or a union? In the first case, seemingly innocent usages will result in undefined behavior:

```
void foo();
function_ref<void()> f{&foo};
f(); // Undefined behavior
```

```
struct foo { void bar(); }
function_ref<void(foo)> f{&foo::bar};
f(foo{}); // Undefined behavior
```

If a union is stored instead, the first usage could be well-formed without any extra overhead (assuming `sizeof(void*) == sizeof(void(*)())`). The second usage could also be made well-formed, but with size overhead as `sizeof(void(C::*)()) > sizeof(void*)`.

Regardless, the exposition-only members should clearly illustrate the outcome of this decision.

Note that if we want the following to compile and be well-defined, a `void(*)()` would have to be stored inside `function_ref`:

```
void foo();  
function_ref<void()> f{foo};  
f();
```

- Should the `function_ref(F&&)` deduction guide take its argument by value instead? This could simplify the wording.

From P0792R0 to P0792R1

Changes

- Removed empty state and comparisons with `nullptr`;
- Removed default constructor;
- Added support for `noexcept` and `const`-qualified function signatures (*these are propagated to `function_ref::operator()`*);
- Added deduction guides for function pointers and arbitrary callable objects with well-formed `&remove_reference_t<F>::operator()`;
- Added two new bullet points to “Open questions”;
- Added “Example implementation”;
- Added “Feature test macro”;
- Removed `noexcept` from constructor and assignment.

Semantics: pointer versus reference

option 1

function_ref, non-nullable, not default constructible

option 2

function_ptr, nullable, default constructible

We want 1 and 2

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1 2 8 3 6

ref vs ptr

SR R N P SP

6 5 2 5 0

The poll above clearly shows that the desired direction for `function_ref` is towards a *non nullable, non default-constructible* reference type. This revision (P0792R2) removes the “empty state” and default constructibility from the proposed `function_ref`. If those semantics are required by users, they can trivially wrap `function_ref` into an `std::optional<function_ref<...>>`.

target and target_type

We want target and target-type (consistent with `std::function`) if they have no overhead

Unanimous consent

We want target and target-type (consistent with `std::function`) even though they have overhead

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I am not sure whether `target` and `target_type` can be implemented without introducing overhead. I seek the guidance of the committee or any interested reader to figure that out. If they require overhead, I agree with the poll: they will be left out of the proposal.

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Overview

Since the advent of C++11 writing more functional code has become easier: functional programming patterns and idioms have become powerful additions to the C++ developer's toolbox. **"Higher-order functions"** are one of the key ideas of the functional paradigm - in short, they are functions that take functions as arguments and/or return functions as results.

The need of referring to an existing `Callable` object comes up often when writing functional C++ code, but the Standard Library unfortunately doesn't provide a flexible facility that allows to do so. Let's consider the existing utilities:

- **Pointers to functions** are only useful when the entity they refer to is stateless (*i.e. a non-member function or a capture-less lambda*), but they are cumbersome to use otherwise. Fully supporting the `Callable` concept requires also explicitly dealing with **pointers to member functions** and **pointers to data members**.
- **`std::function`** seamlessly works with `Callable` objects, but it's a *"general-purpose polymorphic function wrapper"* that may introduce unnecessary overhead and that **owns** the `Callable` it stores. `std::function` is a great choice when an owning type-erased wrapper is required, but it's often abused when its ownership semantics and its flexibility are not required.
 - Note that when `std::function` is constructed/assigned with a `std::reference_wrapper` to a `Callable`, it has reference semantics.
 - Another limitation of `std::function` is the fact that the stored `Callable` must be `CopyConstructible`.

- **Templates** can be used to avoid unnecessary costs and to uniformly handle any `Callable` object, but they are hard to constrain to a particular signature and force code to be defined in headers.

This paper proposes the introduction of a new `function_ref` class template, which is akin to `std::string_view`. This paper describes `function_ref` as a **non-owning lightweight wrapper** over any `Callable` object.

Motivating example

Here's one example use case that benefits from *higher-order functions*: a `retry(n, f)` function that attempts to synchronously call `f` up to `n` times until success. This example might model the real-world scenario of repeatedly querying a flaky web service.

```
struct payload { /* ... */ };

// Repeatedly invokes `action` up to `times` repetitions.
// Immediately returns if `action` returns a valid `payload`.
// Returns `std::nullopt` otherwise.
std::optional<payload> retry(std::size_t times, /* ????? */ action);
```

The passed-in action should be a `Callable` which takes no arguments and returns `std::optional<payload>`. Let's see how `retry` can be implemented with various techniques:

- Using *pointers to functions*:

```
std::optional<payload> retry(std::size_t times,
                           std::optional<payload>(*action)())
{
    /* ... */
}
```

(on godbolt.org) (<https://godbolt.org/g/UQbZYp>)

- **Advantages:**

- Easy to implement: no need to use a template or any explicit constraint (e.g. `std::enable_if_t<...>`). The type of the pointer specifies exactly which functions can be passed, no extra constraints are required.
- Minimal overhead: no allocations, no exceptions, and `action` is as big as a pointer.
 - Modern compilers are able to completely inline the call to `action`, producing optimal assembly.

- **Drawbacks:**

- This technique doesn't support stateful `Callable` objects.

- Using a template:

```
template <typename F>
auto retry(std::size_t times, F&& action)
    -> std::enable_if_t<std::is_invocable_r_v<std::optional<payload>, F&&>,
        std::optional<payload>>
{
    /* ... */
}
```

(on godbolt.org) (<https://godbolt.org/g/AGikkz>).

- **Advantages:**

- Supports arbitrary Callable objects, such as stateful closures.
- Zero-overhead: no allocations, no exceptions, no indirections.

- **Drawbacks:**

- Harder to implement and less readable: users must use `std::enable_if_t` and `std::invocable_r_v` to ensure that `action`'s signature is properly constrained.
- `retry` must be defined in a header file. This might be undesirable when trying to minimize compilation times.

- Using `std::function`:

```
std::optional<payload> retry(std::size_t times,
                            std::function<std::optional<payload>()> action)
{
    /* ... */
}
```

(on godbolt.org) (<https://godbolt.org/g/t9FH9b>).

- **Advantages:**

- Supports arbitrary Callable objects, such as stateful closures.
- Easy to implement: no need to use a template or any explicit constraint. The type fully constrains what can be passed.

- **Drawbacks:**

- Unclear ownership semantics: `action` might either own the stored Callable, or just refer to an existing Callable if initialized with a `std::reference_wrapper`.
- Can potentially have significant overhead:

- Even though the implementation makes use of SBO (*small buffer optimization*), `std::function` might allocate if the stored object is large enough. This requires one extra branch on construction/assignment, one potential dynamic allocation, and makes action as big as the size of the internal buffer.
 - If the implementation doesn't make use of SBO, `std::function` will always allocate on construction/assignment.
 - Modern compilers are not able to inline `std::function`, often resulting in very poor assembly compared to the previously mentioned techniques.
 - Mandatory use of exceptions: `std::function` might throw if an allocation fails, and throws `std::bad_function_call` if it's invoked while unset.
- Using the proposed `function_ref`:

```
std::optional<payload> retry(std::size_t times,
                           function_ref<std::optional<payload>()> action)
{
    /* ... */
}
```

(on godbolt.org) (<https://godbolt.org/g/DvWKVH>)

- **Advantages:**

- Supports arbitrary `Callable` objects, such as stateful closures.
- Easy to implement: no need to use a template or any constraint. The type fully constrains what can be passed.
- Clear ownership semantics: `action` is a **non-owning** reference to an existing `Callable`.
- Small overhead: no allocations, no exceptions, and `action` is as big as two pointers.
 - Modern compilers are able to completely inline the call to `action`, producing optimal assembly.

Impact on the Standard

This proposal is a pure library extension. It does not require changes to any existing part of the Standard.

Alternatives

The only existing viable alternative to `function_ref` currently is `std::function + std::reference_wrapper`. The Standard guarantees that when a `std::reference_wrapper` is used to construct/assign to a `std::function` no allocations will occur and no exceptions will be thrown.

Using `std::function` for non-owning references is suboptimal for various reasons.

1. The ownership semantics of a `std::function` are unclear - they change depending on whether or not the `std::function` was constructed/assigned with a `std::reference_wrapper`.

```
void foo(std::function<void()> f);  
// `f` could be referring to an existing Callable, or could own one.  
  
void bar(function_ref<void()> f);  
// `f` unambiguously is a non-owning reference to an existing Callable.
```

2. This technique doesn't work with temporaries. This is a huge drawback as it prevents stateful temporary lambdas from being passed as callbacks.

```
void foo(std::function<void()> f);  
  
int main()  
{  
    int x = 0;  
    foo(std::ref([&x]{ ++x; })); // does not compile  
}
```

([on godbolt.org](https://godbolt.org/g/DPQ7ku)) (<https://godbolt.org/g/DPQ7ku>)

The code above doesn't compile, as `std::ref` only accepts non-const lvalue references (*additionally, `std::cref` is explicitly deleted for rvalue references*). Avoiding the use of `std::ref` breaks the guarantee that `f` won't allocate or throw an exception on construction.

3. `std::function` is harder for compilers to optimize compared to the proposed `function_ref`. This is true due to various reasons:

- `std::function` can allocate and/or throw exceptions on construction and/or assignment.
- `std::function` might use SBO, which could require an additional branch during construction/assignment, make inlining more difficult, and unnecessarily increase memory usage.

Rough benchmarks comparing the generated assembly of a `std::function` parameter and a `function_ref` parameter against a `template` parameter show that:

- `std::function`, on average, generates approximately 5x more assembly than a template parameter.
- `function_ref`, on average, generates approximately 1.5x more assembly than a template parameter.

A description of the benchmarking techniques used and the full results can be found on my article “*passing functions to functions*”¹.

Synopsis

```
namespace std
{
    template <typename Signature>
    class function_ref
    {
        void* object; // exposition only

        R(*erased_function)(Args...) qualifiers; // exposition only
        // `R`, `Args...`, and `qualifiers` are the return type, the parameter-type-list,
        // and the sequence "cv-qualifier-seq-opt noexcept-specifier-opt" of the function
        // type `Signature`, respectively.

    public:
        constexpr function_ref(const function_ref&) noexcept = default;

        template <typename F>
        constexpr function_ref(F&&);

        constexpr function_ref& operator=(const function_ref&) noexcept = default;

        template <typename F>
        constexpr function_ref& operator=(F&&);

        constexpr void swap(function_ref&) noexcept;

        R operator()(Args...) qualifiers;
        // `R`, `Args...`, and `qualifiers` are the return type, the parameter-type-list,
        // and the sequence "cv-qualifier-seq-opt noexcept-specifier-opt" of the function
        // type `Signature`, respectively.
    };

    template <typename Signature>
    constexpr void swap(function_ref<Signature>&, function_ref<Signature>&) noexcept;

    template <typename R, typename... Args>
    function_ref(R (*)(Args...)) -> function_ref<R(Args...)>;

    template <typename R, typename... Args>
    function_ref(R (*)(Args...) noexcept) -> function_ref<R(Args...) noexcept>;

    template <typename F>
```

```
function_ref(F&&) -> function_ref<see below>;  
}
```

The template argument `Signature` shall be a non-volatile-qualified function type.

<functional> header

Add the following to [functional.syn]:

```
namespace std  
{  
    // ...  
  
    template <typename Signature> class function_ref;  
  
    template <typename Signature>  
    constexpr void swap(function_ref<Signature>& lhs, function_ref<Signature>& rhs)  
    noexcept;  
  
    // ...  
}
```

Specification

```
constexpr function_ref(const function_ref& rhs) noexcept = default;
```

- *Effects*: constructs a `function_ref` referring to the same callable `rhs` refers to.

```
template <typename F>  
constexpr function_ref(F&& f);
```

- *Requires*: none of the following must hold:
 - `f` is a null function pointer value.
 - `f` is a null member pointer value.
 - `remove_cvref_t<F>` is an instance of the function class template, and `!f`.

- *Effects*: constructs a `function_ref` referring to `f`.
- *Remarks*: This constructor shall not participate in overload resolution unless `is_same_v<remove_cvref_t<F>, function_ref>` is false and:
 - If `Signature` is marked `noexcept`: `is_nothrow_invocable_r_v<R, cv-qualifiers F&, Args...>`;
 - Otherwise: `is_invocable_r_v<R, cv-qualifiers F&, Args...>`.

Where `R`, `Args...`, and `cv-qualifiers` are the *return type*, the *parameter-type-list*, and the sequence “*cv-qualifier-seq-opt*” of the function type `Signature`, respectively.

```
constexpr function_ref& operator=(const function_ref& rhs) noexcept = default;
```

- *Postconditions*: `*this` refers to the same callable `rhs` refers to.
- *Returns*: `*this`.

```
template <typename F>
constexpr function_ref& operator=(F&&);
```

- *Requires*: none of the following must hold:
 - `f` is a null function pointer value.
 - `f` is a null member pointer value.
 - `F` is an instance of the function class template, and `!f`.
- *Remarks*: This function shall not participate in overload resolution unless `is_same_v<remove_cvref_t<F>, function_ref>` is false and:
 - If `Signature` is marked `noexcept`: `is_nothrow_invocable_r_v<R, cv-qualifiers F&, Args...>`;
 - Otherwise: `is_invocable_r_v<R, cv-qualifiers F&, Args...>`.
- *Postconditions*: `*this` refers to `f`.
- *Returns*: `*this`.

```
constexpr void swap(function_ref& rhs) noexcept;
```

- *Effects*: exchanges the values of `*this` and `rhs`.

```
R operator()(Args... xs) qualifiers;
```

- *Effects*: equivalent to `return INVOKE<R>(f, std::forward<Args>(xs)...);`, where `f` is the callable object referred to by `*this`, qualified with the same *cv-qualifiers* as the function type `Signature`.

```
template <typename F>  
function_ref(F&&) -> function_ref<see below>;
```

- *Remarks*: This deduction guide participates in overload resolution only if `&remove_reference_t<F>::operator()` is well-formed when treated as an unevaluated operand. In that case, if `decltype(&remove_reference_t<F>::operator())` is of the form `R(G::*)(A...) qualifiers` for a class type `G`, then the deduced type is `function_ref<R(A...) qualifiers>`, where `qualifiers` is the sequence “cv-qualifier-seq-opt noexcept-specifier-opt” of the function type `F`.

```
template <typename Signature>  
constexpr void swap(function_ref<Signature>& lhs, function_ref<Signature>& rhs) noexcept;
```

- *Effects*: equivalent to `lhs.swap(rhs)`.

Feature test macro

I propose the feature-testing macro name `__cpp_lib_function_ref`.

Example implementation

An example implementation is available [here on GitHub](https://github.com/SuperV1234/Experiments/blob/master/function_ref.cpp) (https://github.com/SuperV1234/Experiments/blob/master/function_ref.cpp).

Existing practice

Many facilities similar to `function_ref` exist and are widely used in large codebases. Here are some examples:

- The `llvm::function_ref`² class template is used throughout LLVM. A quick GitHub search on the LLVM organization reports hundreds of usages both in `llvm` and `clang`³.
- Facebook’s Folly libraries⁴ provide a `folly::FunctionRef`⁵ class template. A GitHub search shows that it’s used in projects `proxygen` and `fbthrift`⁶.
- GNU’s popular debugger, `gdb`⁷, uses `gdb::function_view`⁸ throughout its code base. The documentation in the linked header file⁹ is particularly well-written and greatly motivates the need for this facility.

Additionally, combining results from GitHub searches (*excluding “llvm” and “folly”*) for “`function_ref`”¹⁰, “`function_view`”¹¹, “`FunctionRef`”¹², and “`FunctionView`”¹³ roughly shows more than 2800 occurrences.

Possible issues

Accepting temporaries in `function_ref`’s constructor is extremely useful in the most common use case: using it as a function parameter. E.g.

```
void foo(function_ref<void()>);

int main()
{
    foo([]{ });
}
```

(*on wandbox.org*) (<https://wandbox.org/permlink/BPtPeQtErPGj4X7>)

The usage shown above is completely safe: the temporary closure generated by the lambda expression is guaranteed to live for the entirety of the call to `foo`. Unfortunately, this also means that the following code snippet will result in *undefined behavior*:


```
int main()
{
    function_ref<void()> f{[]{} };
    // ...
    f(); // undefined behavior
}
```

(on wandbox.org) (<https://wandbox.org/permlink/cQPEX2sKjCQjgIki>)

The above closure is a temporary whose lifetime ends after the `function_ref` constructor call. The `function_ref` will store an address to a “dead” closure - invoking it will produce undefined behavior ¹⁴. As an example, `AddressSanitizer` detects an invalid memory access in this gist ¹⁵. Note that this problem is not unique to `function_ref`: the recently standardized `std::string_view` ¹⁶ has the same problem ¹⁷.

I strongly believe that accepting temporaries is a “necessary evil” for both `function_ref` and `std::string_view`, as it enables countless valid use cases. The problem of dangling references has been always present in the language - a more general solution like Herb Sutter and Neil Macintosh’s lifetime tracking ¹⁸ would prevent mistakes without limiting the usefulness of view/reference classes.

Bikeshedding

The name `function_ref` is subject to bikeshedding. Here are some other potential names:

- `function_view`
- `callable_ref`
- `callable_view`
- `invocable_ref`
- `invocable_view`
- `fn_view`
- `fn_ref`

Acknowledgments

Thanks to **Agustín Bergé**, **Dietmar Kühl**, **Eric Niebler**, **Tim van Deurzen**, and **Alisdair Meredith** for providing very valuable feedback on earlier drafts of this proposal.

Annex: previously open questions

Below are some unanswered questions for which I kindly ask guidance from members of the committee and readers of this paper.

- `function_ref<Signature>`'s signature currently only accepts any combination of `const` and `noexcept`. Should this be extended to include *ref-qualifiers*? This would mean that `function_ref::operator()` would first cast the referenced callable to either an *lvalue reference* or *rvalue reference* (depending on `Signature`'s ref qualifiers) before invoking it. See P0045R1 ¹⁹ and N4159 ²⁰ for additional context.
 - LEWG agreed that `const` and `noexcept` have useful cases, but we could not find enough motivation to include support for *ref-qualified* signatures. Nevertheless, this could be added as a non-breaking extension to `function_ref` in the future.
- Constructing a `std::function<Signature>` from a `function_ref<Signature>` is completely different from constructing a `std::string` from a `std::string_view`: the latter does actually create a copy while the former remains a reference. It may be reasonable to prevent implicit conversions from `function_ref` to `std::function` in order to avoid surprising dangerous behavior.
 - LEWG decided to not prevent `std::function` construction from `std::function_ref` as it would special-case `std::function` and there are other utilities in the Standard Library (and outside of it) that would need a similar change (e.g. `std::bind`).
- `function_ref::operator()` is not currently marked as `constexpr` due to implementation issues. I could not figure a way to implement a `constexpr`-friendly `operator()`. Is there any possibility it could be marked as `constexpr` to increase the usefulness of `function_ref`?
 - We agreed that there is probably no way of currently having a `constexpr` `function_ref::operator()` and that we do not want to impose that burden on implementations.

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10. [https://github.com/search?utf8=%E2%9C%93&q=function_ref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code_\(https://github.com/search?utf8=%E2%9C%93&q=function_ref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code\)](https://github.com/search?utf8=%E2%9C%93&q=function_ref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code_(https://github.com/search?utf8=%E2%9C%93&q=function_ref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code))↵
11. [https://github.com/search?utf8=%E2%9C%93&q=function_view+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code_\(https://github.com/search?utf8=%E2%9C%93&q=function_view+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code\)](https://github.com/search?utf8=%E2%9C%93&q=function_view+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code_(https://github.com/search?utf8=%E2%9C%93&q=function_view+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code))↵
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